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Using Tufts Dental Database to Predict Segmentation Masks

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Abstract

The purpose of this project is to create a workflow that can take in a panoramic dental radiograph as input and output a predicted segmentation mask that shows where the abnormalities could be on the radiograph. All of the data used came from the Tufts Dental Dataset. I created a binary classifier that predicts whether or not a radiograph has abnormalities, and a segmentation model that takes in a radiograph under the assumption that the binary classifier labeled it as abnormal and predicts the associated segmentation mask. The segmentation model achieved an IoU (Intersection over Union) of 0.087, a dice score of 0.149, and a recall of 39.02%. The binary classifier achieved an accuracy of 61.33%, precision of 46.15%, a recall of 82.35%, and an F1 of 0.592. The results show that the models learned, and this project could be a success in terms of machine learning, but the results are not accurate enough to be used in a clinical setting.

Introduction

For this project, I created a workflow that uses a binary classifier followed by a segmentation model to predict where abnormalities exist on input panoramic dental radiographs. All of the data used to train both models came from the Tufts Dental Dataset [1] that contains 1,000 de-identified panoramic dental radiographs. I used three metrics to evaluate the segmentation model and four metrics to evaluate the binary classifier. I made a binary classifier that takes in a radiograph as input and predicts whether or not the radiograph has one or more abnormalities. I also made a segmentation model, which takes in radiographs predicted by the binary classifier to have abnormalities and outputs a segmentation mask that shows where the model predicts where the abnormalities could be. The segmentation model runs under the assumption that the input radiograph has abnormalities.

The data for this project comes from the Tufts Dental Database (Figure 1). [1] The Tufts Dental Database includes 1,000 de-identified panoramic radiographs of Tufts Dental School patients. The radiographs were selected from the electronic patient database (axiUm) at the dental school. The images selected had to be high quality and with minimal or no technical errors. Each radiograph was annotated by an expert and by a student at the Tufts Dental School. In addition to the annotated radiographs, eye-tracking and audio data were collected from both the expert and the student. Radiographs are the images produced when one takes dental x-ray.

Segmentation masks are binary images that show the locations of abnormalities on the radiograph. An abnormal region on a segmentation mask is depicted as white pixels on a black background. For the purposes of the project, a positive pixel is white (abnormal) and a negative pixel is black (normal).

The dataset creators used diagnostic radiography to improve image quality. The image enhancement increased contrast, optimized brightness, improved sharpness, and reduced noise. More specifically, the CLAHE algorithm performed the best for enhancing the images.

Binary classifier: The binary classifier uses an EfficientNet-B0 architecture. For the classifier, I used accuracy, precision, recall, and F1 as well as a confusion matrix (table 1). Accuracy is the percentage of radiographs in the test set that were predicted correctly as normal or abnormal. Precision is the percentage of radiographs that were correctly predicted as abnormal ($\frac{TP}{TP+FP}$). Recall is the percentage of abnormal radiographs that were correctly identified ($\frac{TP}{TP+FN}$). F1 is calculated as $\frac{2*(recall*precision)}{recall+precision}$. Recall and precision have an inverse relationship due to the fact that recall reduces false negatives and precision reduces false positives.

Segmentation model: The segmentation model uses a U-Net CNN architecture with patch-based learning. To evaluate the model, I used IoU, dice, and recall. IoU is calculated by dividing the intersection (number of white pixels that exist in both the predicted segmentation mask and the ground truth) by the union (number of white pixels that are in either the predicted segmentation mask or the ground truth). Dice is calculated by multiplying the intersection by 2 and dividing the product by the total number of white pixels in both images. Dice is "the current *de facto* standard to determine agreement between a reference segmentation and one generated by manual / auto-contouring approaches" (McCullum et. al, 2024)[2] and IoU is a measure of how much two regions in two figures overlap.

The segmentation model achieved an IoU (Intersection over Union) of 0.087, a dice score of 0.149, and a recall of 39.02%. The binary classifier achieved the following confusion matrix: TP = 42, FP = 49, FN = 9, TN = 50. Additionally, the binary classifier has an accuracy of 61.33%, precision of 46.15%, a recall of 82.35%, and an F1 of 0.592. The results show that the models learned, and this project could be a success in terms of machine learning, but the results are not accurate enough to be used in a clinical setting.

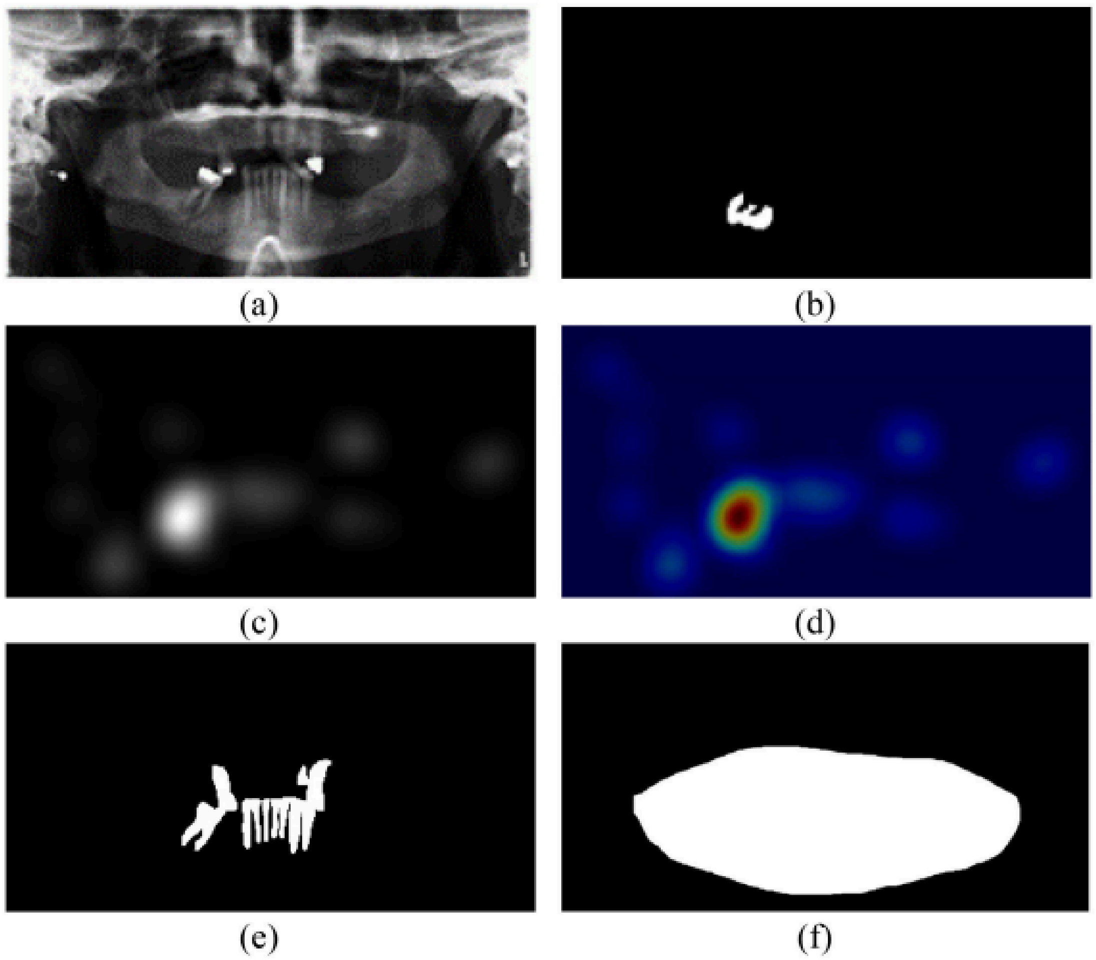


Fig. 3. Illustration of the different imaging components. (a) panoramic radiograph, (b) segmented mask outlining the abnormality, (c) grayscale eye-tracking gaze plot, (d) color quantized eye-tracking gaze plot (e) teeth mask, (f) maxillomandibular region of interest.

Figure 1. Source: Panetta, Karen. *Tufts Dental Database: A Multimodal Panoramic X-Ray Dataset for Benchmarking Diagnostic Systems* [1]

Initial Problem Proposal and Revisions

Initial problem proposal: Can data collected from Tufts Dental School be used to predict whether or not a patient will need a root canal based on the depth of a patient's cavity?

Problem revisions: Throughout this project, there were two problem revisions. When I realized that the Tufts Dental Database does not have any data on depth of cavities, I decided to switch the problem to using radiographs to predict gaze heatmaps. This was because the gaze heatmaps are part of what make the Tufts Dental Dataset unique. Gaze heatmaps are RGB images that show where the individual looking at the radiograph focused their attention. An example of a gaze heatmap is image (d) in figure 1. However, early in the spring semester I realized that predicting the gaze heatmaps does not have enough relevance for the purpose of this project. It would be useful for dental students and experts, but it would not have been useful for patients. Therefore, I decided as my final problem revision to use radiographs to predict segmentation masks. Considering all of the data available in the Tufts Dental Database, this problem offered the most linearity of any other problem I could have chosen. A model that predicts where abnormalities are on a radiograph is useful for dental experts and students and it can also more easily capture the interest of patients. Patients are generally more interested in whether or not something is wrong with their teeth than where the dentist should focus their attention.

Prototypes and Proof of Concept: During the fall semester, I analyzed the gaze heatmaps to see what parts of the gaze heatmaps could potentially be classified as noise. Since, as described by Christopher Bishop in *Pattern Recognition and Machine Learning* [3], noise is random data that corrupts observations, I wanted to explore if there was any part of the gaze heatmaps that I could consider to be noise. I applied a gaussian filter with a sigma value of 20 on four gaze heatmaps from the Tufts Dental Dataset, which had no effect on the images. I also applied a threshold by controlling which pixels made it to the output based on hue, saturation, and brightness. The brightness threshold value used was 0.38. The applied threshold removed all of the light blue regions. Light blue regions on a gaze heatmap specify where the individual looked at the associated radiograph for a short period of time.

Using the PIL library in Python, I was able to determine that each image has 840x1615 pixels which will make it consistent to load the images into a CNN. I later realized I could verify this by opening the images on my local machine and the image dimensions were listed right under the image.

An interesting characteristic of the data I noticed is that in segmentation masks that are empty (meaning there is no abnormality), the corresponding gaze heatmap generally has several gaze circles where a gaze circle is a point on the radiograph where the student or expert looked for a non-negligable amount of time. A gaze heatmap with several gaze circles indicates that it is possible the individual looking at the radiograph may not have one particular spot of interest if the radiograph has no abnormalities.

When I started working with the segmentation masks, one thing that I noticed was that the .JPG format of the masks caused the perimeter of white regions to contain grey pixels. Converting the images to .png format prior to training the models helped avoid compression-related grey pixels near the perimeter. These grey pixels are an artifact of jpeg compression, which the lossless .png format does not suffer from.

Figure 2: This JPG image is directly from the Tufts Dental Database. Grey pixels line the border of the white region[1].

Figure 3: This image is a higher resolution PNG lossless. It is figure 2 converted to a PNG image.

Experimental design

I trained two different models, a binary classifier and a segmentation model. The binary classifier predicts whether or not a radiograph's associated mask has abnormalities. Using the binary classifier at inference time, I can allow the segmentation model to operate entirely on radiographs with anomalies. Because the segmentation model only ever sees images the classifier has already flagged as abnormal, I trained it exclusively on the 339 abnormal radiographs and avoided the need to heavily weight the Binary Cross Entropy (BCE) loss to compensate for image-level classification imbalance. This allows our hyperparameters to be better tuned to the positive cases rather than mitigating non-anomaly features such as BCE weight. The final workflow is as follows: if the binary classifier identifies a radiograph with abnormalities, the radiograph is segmented via the segmentation model, which outputs the predicted mask (figure 4).

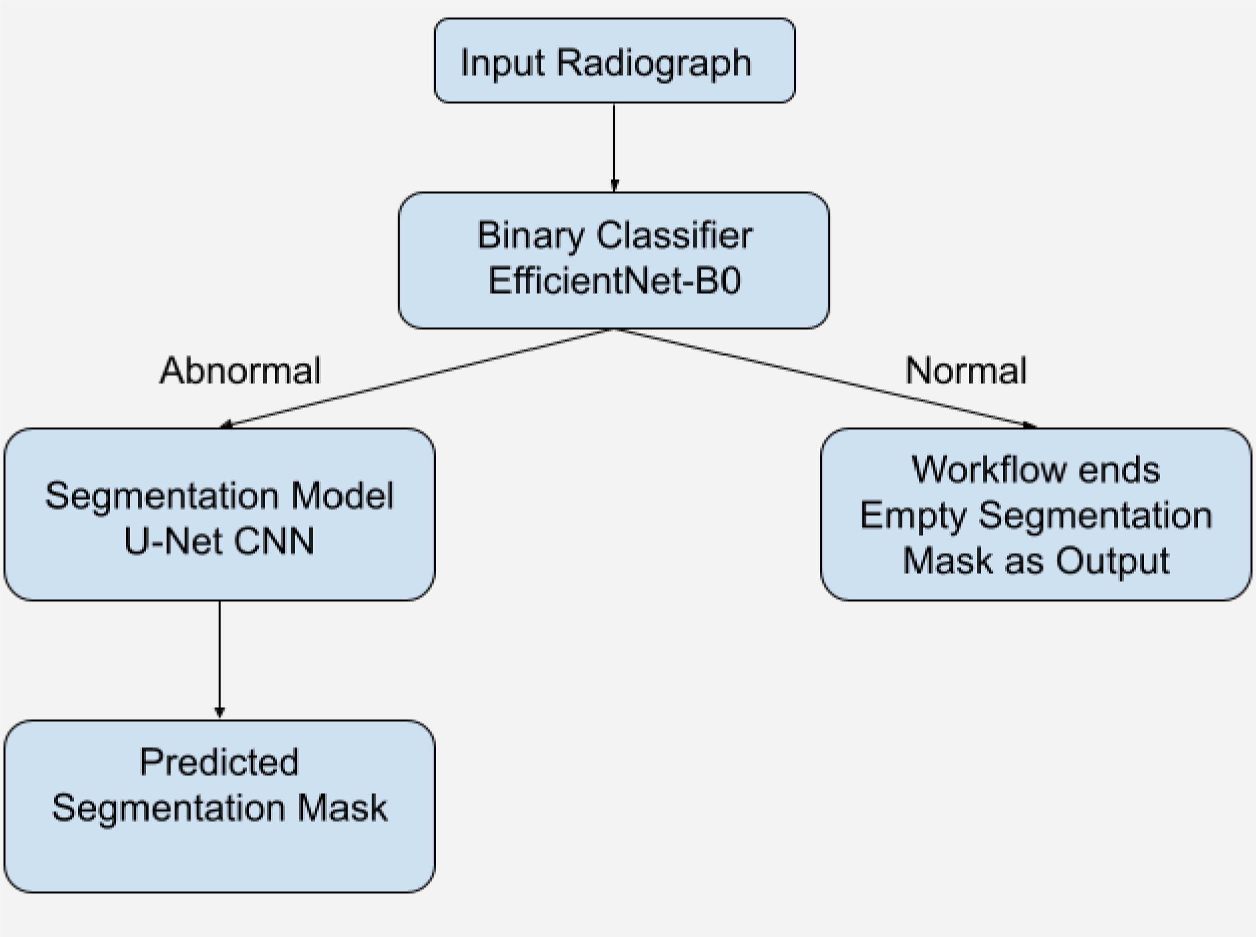


Figure 4: Experimental design workflow. The binary classifier takes in a radiograph, and if it predicts that the radiograph has anomalies, the segmentation model takes the radiograph as input and outputs a segmentation mask that predicts where the anomalies could be. If the binary classifier predicts that the input radiograph has no anomalies, a segmentation mask with no white pixels gets produced.

Binary Classification Model:

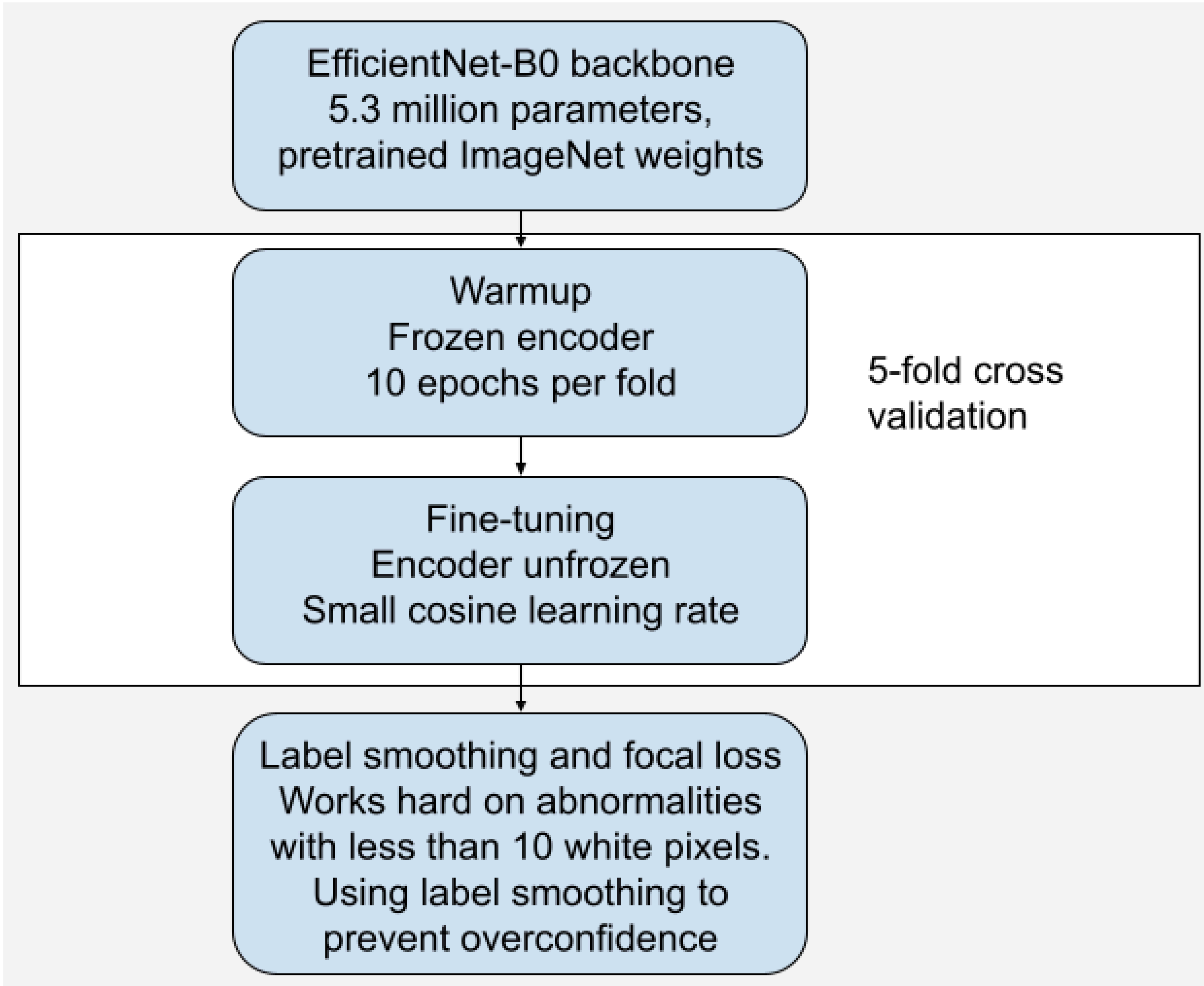


Figure 5: Binary classifier architecture diagram. The model takes in the training and validation data to perform 5-fold cross validation. A threshold sweep is performed on the validation set to determine how confident the model should be in order to label a radiograph as abnormal.

The binary classifier uses an EfficientNet-B0 architecture, which makes it easiest for the model to learn since it only has 5.3 million parameters (Figure 5). Mistry et. al [4] used EfficientNet-B0 in order to identify whether or not images are AI generated. The training process utilizes 5-fold cross validation where the encoder is frozen for the first 10 epochs in each fold. When an encoder is frozen, it means the weights of the encoder do not get updated. Freezing the encoder helps prevent the model from overfitting because it gives the weights fewer opportunities to update. Once the encoder is unfrozen, I implemented a very small learning rate that grew very slowly to prevent overfitting. Additionally, the model employs a focal loss function which forces the model to learn more about the more difficult radiographs where the abnormality might be less than 10 white pixels (Figure 5). The focal loss function also uses label smoothing which stops the classifier from being too confident.

To train and test the binary classifier, I created an 68/17/15 training/validation/testing split in both the masks and radiographs making sure the training and testing folders each contained 33% to 34% images with abnormalities. Additionally, I performed a horizontal flip on all of the training images to achieve a total of 1,360 masks and 1,360 radiographs for training and validation. While in theory mouths are symmetric, there are many radiographs in the Tufts Dental Dataset where the patient was missing 1 or more teeth meaning the mouth of that patient was not symmetric. Therefore, by performing data augmentation, I am able to create unique radiographs to add to the data.

In order to determine the best model achieved during 5-fold cross validation, the best model is chosen based on F1 score to balance precision and recall Once the best model was chosen from the cross validation, a threshold sweep was performed with the following thresholds: 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70, 0.75, 0.80. In this case, threshold is a value that tells the model how confident it has to be to label a radiograph as abnormal. The best threshold for the binary classifier is 0.45, meaning the model has to be at least 45% confident to label a radiograph as abnormal. With a threshold of 0.45, the best model has an accuracy of 61.33%, precision of 46.15%, recall of 82.35%, and F1 score of 0.592. There are 42 true positives, 49 false positives, 9 false negatives, and 50 true negatives. For the purpose of this being used as a diagnostic tool, a low number of false negatives was prioritized.

Segmentation Model:

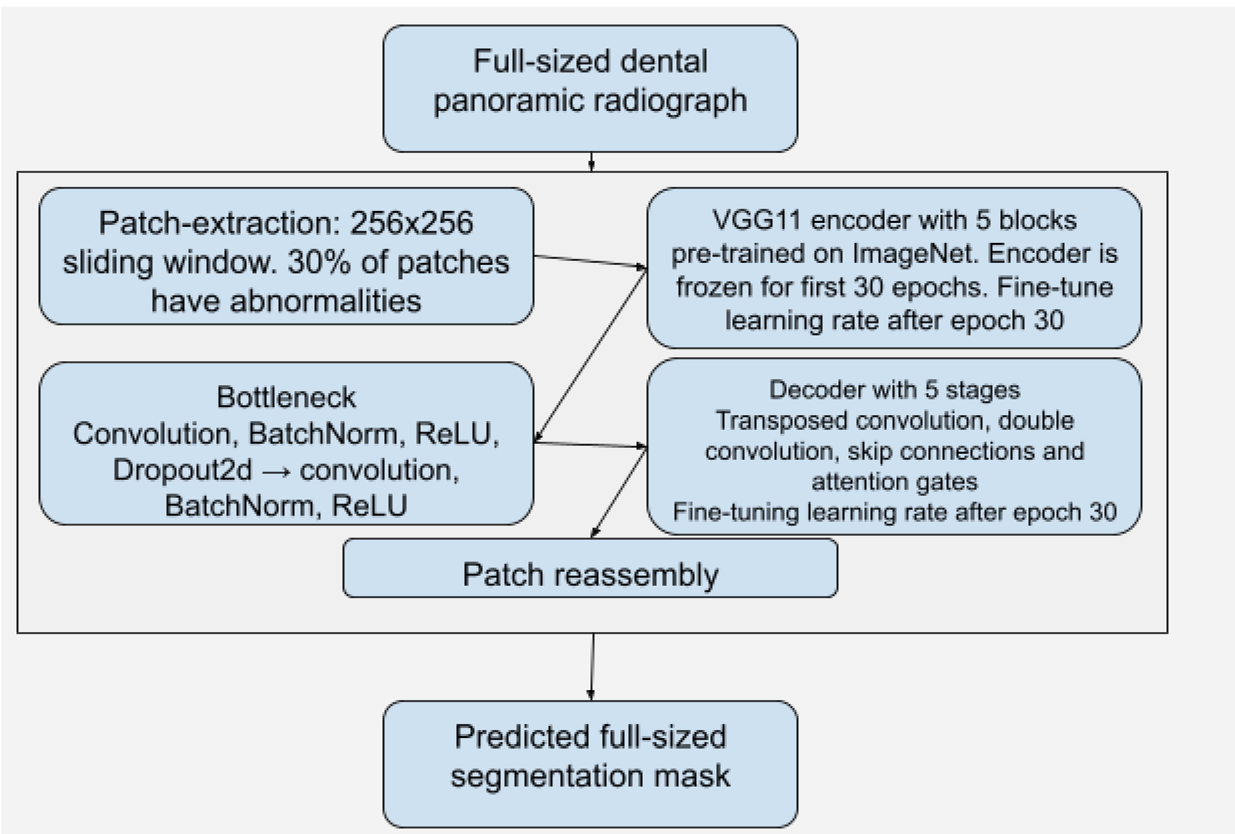


Figure 6: Segmentation model architecture diagram. The model takes in the training and validation data, and uses patch-based learning to learn from the images. After 128 epochs, a threshold sweep is done on the validation set to determine how confident the model should be to label a pixel as abnormal.

I trained a convolutional neural network (CNN) on the radiographs and the segmentation masks. The goal was to train a CNN that was pre-trained on ImageNet which has just over 14 million images, so when the model is fed a radiograph it can output a predicted segmentation mask. I used a pre-trained CNN because the Tufts Dental Dataset does not have enough data to train an entire CNN from scratch, so I needed to use a CNN that was already trained on a wide variety of images. My focus is on CNNs because pre-trained CNNs have been widely used in dentistry for the diagnosis of various dental anomalies (Panetta et al., 2022)[1]. Muresan [5] uses both image processing and deep learning to detect teeth and dental problems in panoramic x-rays. Muresan specifies using CNNs and claims "convolutional neural networks (CNN) are preferred solutions for medical imaging analysis, and have been employed in many clinical fields" (Muresan 2020). Stember [6] studied whether or not medical segmentation masks that were generated using eye tracking are equivalent to medical segmentation masks made using hand annotation. In this specific study, Stember used eye tracking masks to train a CNN. Deep learning by way of CNNs "has enabled the segmentation of anatomic structures previously difficult to segment with older methods of thresholding, region/boundary-based approaches, or hybrid techniques such as combining models and graph-search algorithms" (Stember et al., 2019). Stember [6] does not mention any denoising that they had to do to the images prior to running them through the CNN.

CNNs work via 3 types of layers: a convolutional layer, a pooling layer, and a fully-connected layer. The convolutional layer specifically consists of 3 components: "input data, a filter and a feature map" (IBM)[7]. When a CNN is fed a colored image, it uses length, width, and depth as the 3 dimensions to process the image. After the first layer of the CNN, I use the word 'depth' to refer to the number of feature maps.

A kernel is used to analyze the images and check for features. [8] A kernel is a very small matrix, where the size of the matrix determines how much data is considered at once. The sliding window method is applied to a kernel. [9] In the sliding window method, the kernel is applied to every region of the image. For each location where the kernel is, I multiply the kernel weights with the pixel values and then I add the products to get a value. Gradient descent and back propagation are used to change the kernel weights. Kernels output a feature map which is generated using the sliding window method and shows the features that the kernel detected in the data. [7] The entire process of using a kernel to analyze any potential features in an image is called convolution because the kernel performs mathematical convolution every time it slides over new data.

There were some challenges to address when considering the model's implementation. "A key limitation to the advancement of medical DL segmentation is obtaining abundant quantities of accurately curated imaging data for effective DL training. Specifically, what is needed to build successful CNNs is a large number and variety of images, with corresponding expert-annotated masks" (Stember et al., 2019)[6]. Our data has about 1,000 radiographs and segmentation masks. This may not have been enough images to train a deep neural network from scratch. Since our dataset is not large enough to train a CNN from scratch, I used a CNN pre-trained on ImageNet that had already been trained to analyze and gather appropriate features from images [10]. Given the fact that the data is both well-labeled and high quality, a sufficiently tuned pre-trained CNN did not require very much data manipulation for training. The CNN I used was pre-trained using ImageNet. In particular, I focused on using the U-Net architecture with a VGG11 backbone (Figure 6).

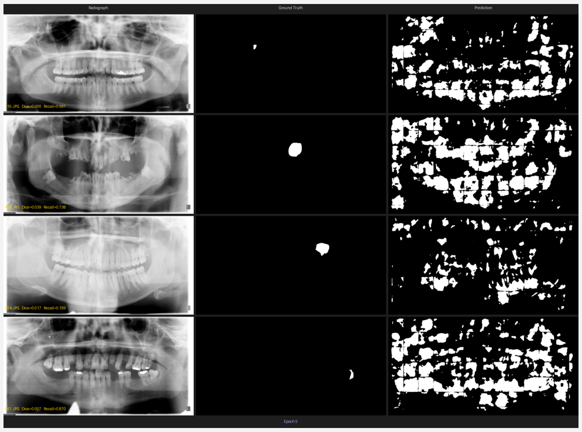
I used a U-Net CNN pre-trained with ImageNet because "the architecture consists of a contracting path to capture context and a symmetric expanding path that enables precise localization" (Ronneberger et al., 2015)[11]. I used a VGG11 backbone pretrained on ImageNet and a head consisting of the decoder and the output layer as these are the "task specific [parts] of the model" (Moazeni, 2025)[12].

Attention gates are employed in an attempt to remove as much noise as possible. Attention gates are a part of skip connections, where the weights of the attention gates are updated from backpropagation and are used to determine which aspects of the feature maps are most important. Skip connections act as a way to pass information from the encoder to the decoder without compression so no important information gets lost. Additionally, I implemented patch-based learning (Figure 6). Patch-based learning is a method commonly used in medical image segmentation where you split an image into patches in order to amplify tiny parts of the image. Sogur and Çerçevik [13] used patch-based learning to feed dental panoramic radiographs through a CNN. They noticed dental panoramic radiographs often get produced inconsistently, for various reasons such as differences in x-ray machines and differences in patient jaws. Therefore, they developed a system that takes in patches of radiographs as input and outputs a binary mask where white regions represent teeth and black regions represent other tissue. During training, a 256x256 patch with an abnormality is cut out of the masks along with 256x256 patches that have no abnormalities. In training, 30% of the patches have abnormalities and 70% do not. The encoder consists of 5 blocks where each block produces a feature map. Then I have the bottleneck which consists of two blocks where the first block has convolution, BatchNorm, ReLU, and Dropout2d and the second block has convolution, BatchNorm, and ReLU. Finally, the decoder uses 5 upsampling stages that consist of transposed convolution followed by a double convolutional block. Each stage uses skip connections and an attention gate to try to reduce noise. During training, the encoder is frozen for the first 30 epochs and then is unfrozen for fine tuning at a learning rate of 5e-5 with the decoder having a learning rate of 1e-4 for the remaining 98 epochs.

The data to train the model consists of 339 radiographs and 339 associated masks. The data was split into training, validation, and testing folders where the training folder contains 232 radiographs and masks, the validation folder contains 57 radiographs and masks, and the testing folder contains 50 radiographs and masks. A horizontal flip was performed on all of the training data in order to double the training dataset since 232 images is too small to train a CNN.

The best model is saved based on validation Intersection over Union (IoU). IoU is a metric that measures how well two images overlap. In our case, it measures how well a predicted mask overlaps with its associated ground truth mask. After the best model is selected, a threshold sweep is done on the validation set using the following values: 0.20, 0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55, 0.60. The best threshold chosen was 0.45. The best model was analyzed using IoU, dice, and recall. I achieved an IoU of 0.0867, a dice of 0.1497, and a recall of 0.3902. Recall was used exclusively on the areas where the abnormalities were supposed to be.

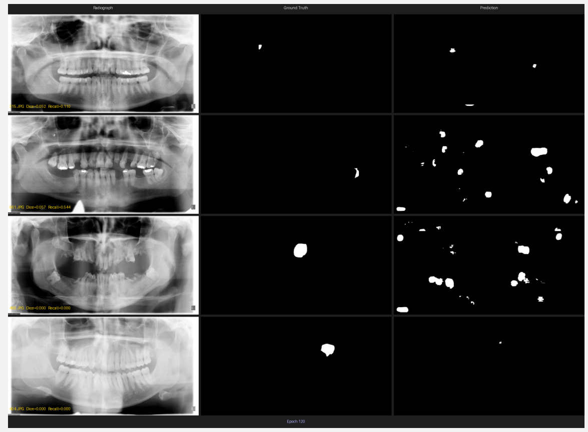
Epoch 5



Radiograph GT Mask Pred Mask

Figure 7: The segmentation model results after 5 epochs. By this point, the model has not learned anything yet. These results are the same as what I was seeing when I trained the segmentation model for the first time on all of the data. Something about these results that I find interesting is that the white regions actually look like full teeth. This could indicate that it is possible that the model started by learning where the teeth are in the radiographs.

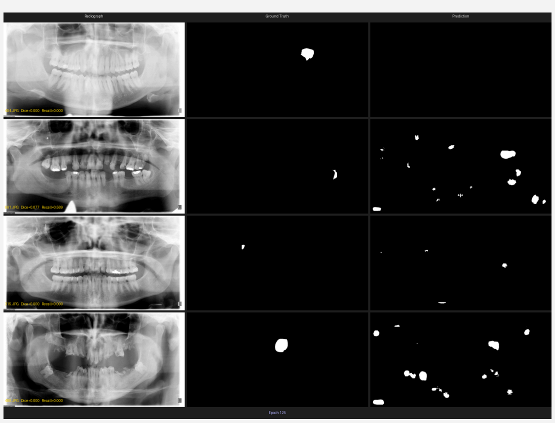
Epoch 120



Radiograph GT Mask Pred Mask

Figure 8: The segmentation model results after 120 epochs. By this point, the model has appeared to learn something as shown by the fact that the regions with the white pixels are a lot more localized by 120 epochs. In fact, the predicted mask at the bottom performed very well with a few white pixels only at the place where the abnormality is on the ground truth mask. That being said, the predicted image second from the bottom is the only image of the 4 that does not appear to have identified the correct location of the abnormality.

Epoch 125



Radiograph GT Mask Pred Mask

Figure 9: The ground truth images were generated after 125 epochs. Interestingly, the predicted image in the top right shows a black rectangle meaning the model after 125 epochs could not identify the location of the abnormality in the radiograph in the top left. That being said, the rest of the predicted segmentation masks look fairly similar to those in figure 8.

Results

Our findings indicate that the models could be considered a success in terms of machine learning, but our models are not successful when you consider them as diagnostic tools. Refer to the introduction section for explanation of the metrics used to evaluate both models.

Predicting segmentation masks from dental panoramic radiographs is a challenging task especially with only 1,000 images. The problem is comparable to finding a needle in a haystack due to the class imbalance. Only 339 of the 1000 radiographs have abnormalities, and of the 339 masks that have abnormalities, 282 consist of less than 1% white pixels and 334 consist of less than 5% white pixels. That being said, as shown by comparing figures 7 and 8, the segmentation model did appear to learn something. However, given an average dice score of 0.149 and IoU of 0.0867, as well as figures 7, 8, and 9, it is clear that the segmentation model tends to predict abnormalities in the wrong regions and sometimes misses abnormalities entirely.

The binary classifier got an accuracy of 61.33%, precision of 46.15%, recall of 82.35%, and F1 of 0.592. The core issue is that there is a tradeoff between recall and precision. Recall reduces false negatives, whereas precision reduces false positives. This means that they have an inverse relationship. In the case of the binary classifier, the goal was to minimize false negatives since a false negative is a missed abnormality. If this model was used to treat patients, it would be detrimental to have a high number of false negatives, but it would be ok to have a high number of false positives because the dentist is able to check to see if an abnormality exists.

True Positives: 42	False Positives: 49
True Negatives: 50	False Negatives: 9

Table 1: Binary classifier confusion matrix. True positive: radiograph has abnormalities and is predicted as having abnormalities. False positive: radiograph has no abnormalities and is predicted as having abnormalities. True negative: radiograph has no abnormalities and is predicted as having no abnormalities. False negative: radiograph has abnormalities and is predicted as having no abnormalities.

Dead Ends and Unsuccessful Experiments

Throughout this project, I have had several dead ends and failed approaches. I experienced failures ranging from using the wrong hardware and running pre-trained models incorrectly to data leakage. While crafting my first attempt, I did not realize how long a neural network can take to run. Therefore, I tried running the program overnight on my machine and over the course of 10 hours it had only completed 1 epoch. I switched to google colab pro +, and trained the model unaware that ImageNet and VGG11 were both trained on exclusively RGB images. Therefore, all of my predicted masks came out completely black. After realizing this error, I simply replicated the single grayscale channel 3 times so the radiographs could be fed into the model, and all of the predicted masks came out with abnormalities (white pixels).

When I realized the issue was that the model was not learning about what abnormalities might look like, I did research and discovered patch-based learning. While the patch-based learning appeared to be helping, I discovered massive data leakage issues as I did not create separate training, testing, and validation folders beforehand. I thought I had reached a dice score of about 60%, but I realized this was invalid due to the data leakage.

Lessons Learned

Throughout this whole process, I have learned that doing research can teach you so much more than any class. I feel like doing this project has taught me more than every class I have taken combined. I learned that noise is not always going to be obvious, as it could very easily be elusive. I also learned that fine-tuning a pre-trained CNN for a specific goal is computationally expensive where 128 epochs can easily take 4.5 to 5 hours.

This project taught me how to use a GPU, which is a very good skill for a data scientist. Prior to GPU access, I attempted to run 50 epochs on my local machine and half an epoch was complete after 10 hours.

If I started this project again, I would begin by first making both a binary classifier and a segmentation model. I would explore what the simplest model is that I could possibly use. I would explore simpler models because I hypothesize my U-Net CNN architecture might have too many parameters considering the limited data available. A model with too many parameters compared to the amount of data available is likely to overfit. I used a U-Net CNN architecture for this project because I already understood it from when I was working with the gaze heatmaps, but I do not know if it was the best architecture for this project.

Additionally, I would explore incorporating other parts of the data into both of the models. I know the dataset contains text and audio annotations from both the students and experts, and I think it would be interesting to incorporate these into the models. That being said, I think the most effective addition would be to incorporate the teeth masks to help guide the model towards where the teeth actually are on the radiographs.

Open Problems for Future Work

There are several ways I would like to see these models improved, from using different architectures to introducing one or more additional modalities of the data.

I would explore to see if there are architectures with fewer parameters that might be more successful when it comes to both models. I would like to see if there is a simpler CNN architecture that can be used for the binary classifier that has fewer than 5.3 million parameters. I would also like to explore using LeNet-5 for the segmentation model to see if that could improve the test recall and/or test dice score.

Additionally, I would like to explore whether incorporating different modalities of the data can have a positive impact on the performance of one of or both of the models. I would first try to find some way to implement the teeth masks so the models know where the teeth are in the radiographs.

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